

Task 8: Train-Test Split Report

Mobile Price Prediction Classification Project | 19 September 2026

1. Business Requirement

The mobile-phone company requires a machine-learning solution that classifies phones into price categories using technical specifications such as RAM, storage, battery capacity, camera quality, screen resolution, processor features, connectivity, and other hardware characteristics.

2. Task Objective

Task 8 prepares the model-ready dataset for Random Forest classification by separating it into independent training and testing datasets. The split must be reproducible, preserve all encoded features, retain the target label, and maintain a similar distribution of price categories in both subsets.

3. Methodology

The input file was `ml_ready_dataset.csv`, produced during feature engineering and encoding. `Price_Category_Encoded` was used as the target column, while the remaining 26 columns were used as model features. An 80/20 split was created with scikit-learn's `train_test_split` function using stratification and `random_state=42`.

4. Split Summary

Measure	Result
Total records	1,017
Training records	813 (80%)
Testing records	204 (20%)
Feature columns	26
Target column	Price_Category_Encoded
Split strategy	Stratified
Random state	42

5. Target-Class Distribution

Price category	Training count	Training %	Testing count	Testing %
Below 15,000	230	28.3%	58	28.4%
15,000-29,999	294	36.2%	74	36.3%
30,000 and above	289	35.5%	72	35.3%

The close percentages between the training and testing subsets confirm that stratification preserved the representation of all three price categories. This supports fair model training and evaluation.

6. Deliverables

Training dataset: training_dataset.csv (813 records) Testing dataset: testing_dataset.csv (204 records)

Implementation: train_test_split.py

7. Conclusion

Task 8 is complete. The model-ready mobile-phone data has been divided into reproducible, stratified training and testing datasets. Both files contain the same 26 encoded feature columns and the Price_Category_Encoded target, making them ready for Task 9: Random Forest Classification.

Generated from the project datasets by generate_task_8_report.py.